

ShadowLearn User Manual

Welcome to ShadowLearn!

Overview

ShadowLearn is your ultimate companion to conquer any foreign language using the **Shadowing** technique (Listen & Repeat) combined with smart interactive exercises and AI Assistance.

This guide will walk you through setting up and maximizing the application’s features.

1. Initial Setup (For Beginners)

When you launch the app for the **first time**, you will see a **Setup** screen asking for **API Keys** and a **PIN**. This only happens once.

What are API Keys?

They are “keys” that allow the app to connect with Artificial Intelligence (AI) models to power Translation, Speech Recognition, and Pronunciation features.

Key Name	Main Role	Importance
OpenRouter API Key	Translates, explains grammar, and chats with you.	Required
Google Gemini API Key	Powers live voice conversations in Speak with AI.	Required for Speak with AI

Absolute Security: All keys are encrypted and **only saved on your computer**. The app never sends them to our servers.

How to Get API Keys

OpenRouter API Key (Translation & Explanation)

- See tutorial here: <https://app.tango.us/app/workflow/Create-API-Key-in-OpenRouter-95b3cf6d85df49b48ad728d93c8d5b76>

Google Gemini API Key (Speak with AI)

Video walkthrough: <https://app.tango.us/app/workflow/Create-Google-AI-Studio-API-Key-ac110f276682495784c9fa2998323bee>

1. Go to aistudio.google.com and sign in with your Google account
2. Click **Get API key** in the left sidebar
3. Click **Create API key** → select **Create API key in new project** → copy the key (starts with AIza...)
4. Still inside AI Studio, open **Billing** from the left menu → click **Add payment method** → enter your **Visa / Mastercard** details and confirm. **Do not use the free trial** — it is unstable and will cause the voice session to drop or fail. Paid billing is required for reliable Gemini Live performance.
5. Open ShadowLearn → **Settings** → paste the key into the **Google Gemini (Realtime)** field and save

 Google AI Studio — API key and billing setup
Google AI Studio — API key and billing setup

Cost: ~300 VNĐ (~\$0.012 USD) per minute of conversation. You only pay when you use it.

Create a PIN

- Create a PIN (minimum 4 characters) and confirm it.
- The PIN encrypts your keys and lesson data while the app is closed.

Important: If you forget your PIN, you will need to reset the app and re-enter all API keys. There is no recovery option. Write your PIN somewhere safe!

 Setup Screen
Setup Screen

2. Daily Use: Unlocking the App

Every time you re-open ShadowLearn, you will see the **Unlock** screen. Simply enter your PIN to access your lessons and data.

| This is normal — it means your data is protected when you're not using the app.

3. Library & Lesson Management

The main dashboard is the **Library**, containing all your created lessons.



Creating a New Lesson

Click the + card or **Create Lesson** button. You have 2 options:

1. **Via YouTube Link :**
 - Paste the video URL into the box.
 - The app will download the audio and generate subtitles automatically.
2. **Upload from Computer :**
 - Upload an audio (.mp3) or video (.mp4) file from your device.

Select your target language (e.g., **English**), then click **Generate**. The lesson enters a processing queue — you can return to the Library to wait (status shown as *Processing*).



Managing Existing Lessons

Each lesson card in the Library supports:

- **Rename** — Click the pencil icon on a card to give it a new name.
- **Delete** — Remove the lesson and all its data permanently.
- **Retry** — If a YouTube lesson failed to process, use Retry to try again.
- **Search** — Use the search bar at the top to filter lessons by title.

4. Learning Interface (Classroom)

Opening a lesson divides the screen into **3 main columns**:



Left Column — Video / Audio Player

Plays the lesson content. Controls include:

- **Play / Pause** button
- **Playback speed**: Choose 0.5×, 0.75×, 1×, 1.25×, or 1.5× — slower speeds are great for beginners
- **Seek bar**: Click anywhere on the timeline to jump to that moment
- **Volume slider**: Adjust audio volume
- **Download** : Save the video/audio file to your computer
- **Rename** : Click the pencil icon next to the lesson title to rename it

Middle Column — Subtitles & Practice

- Displays all segments with **Pinyin**, **Chinese text**, and **translation**.
- **Click any segment** to jump the video to that exact moment.
- **Hover a word** to see its pronunciation and a bookmark icon.
- **Bookmark** : Save any word to your Vocabulary Workbook.
- **Play** : Hear the AI pronounce any segment.
- **Copy** : Copy the segment text to clipboard.
- **Shadow** : Start a Shadowing session beginning from that specific segment.
- **Shadowing button** at the top: Start a full Shadowing session for the lesson.

Right Column — AI Companion (2 Tabs)

The right column is home to your personal AI tutor — an intelligent agent that knows your vocabulary, tracks your progress, and runs interactive exercises directly inside the chat. See [AI Companion ↓](#) for the full guide.

Tab 1 — Chat — Talk to your AI tutor, ask questions, request exercises. **Tab 2 — Workbook** — Browse words saved from this lesson; click any word to jump to it in the transcript. **Study This Lesson** button launches a quiz session.

5. AI Companion — Your Personal Language Tutor

Most language apps give you exercises. ShadowLearn gives you a tutor.

The **AI Companion** lives in the right panel of every lesson. It is not a simple chatbot — it is an intelligent, agentic tutor that **knows who you are**, **remembers you across sessions**, and **takes action** on your behalf. It can read your vocabulary list, check which words you’re due to review, see exactly which sentence you’re studying, and then launch a real interactive exercise — all directly inside the chat.

What Makes It Different

Ordinary chatbot	ShadowLearn AI Companion
Answers questions	Answers questions and acts on them
Forgets you next session	Remembers your preferences, struggles, and goals
Gives generic advice	Adapts to your level and progress in real time
Talks about exercises	Runs exercises directly in the chat

How to Start a Conversation

Just type naturally. Here are ideas to get you going:

Ask about the current sentence: - “What does this sentence mean?” - “Break down the grammar of this line for me.” - “What’s the difference between 了 (le) and 过 (guò)?”

Ask for practice: - “Quiz me on the words I saved from this lesson.” - “Give me a dictation exercise.” - “Can we practice the pronunciation of this sentence?” - “I keep forgetting 尤其 — can we drill it?”

Ask for a progress check: - “How am I doing overall?” - “Show me my accuracy trend.” - “What words should I review today?”

Tell it about yourself: - “I’m a complete beginner, please keep explanations simple.” - “My goal is to pass HSK 3 by the end of the year.” - “I struggle with tones — focus on that when you quiz me.”

The AI will remember everything you tell it, even in future sessions.

Interactive Exercises Right in the Chat

This is the standout feature. When you ask for practice, the AI doesn’t just describe an exercise — it **renders a live, interactive exercise directly inside the conversation**. You complete it, submit your answer, and the AI responds with feedback and updates your progress automatically.

Exercises the AI can launch in chat:

Exercise	What happens
Dictation	An audio clip plays; you type what you hear
Character Writing	A character appears; you trace its stroke order
Pinyin Recall	A character appears; you type its pinyin
Translation	A sentence appears; you translate it
Pronunciation	A sentence appears; you record yourself saying it
Cloze (Fill-in-the-blank)	The AI writes a short story using your words with gaps to fill
Reconstruction	Word chips appear scrambled; you drag them into the right order

The AI chooses which words to include based on what you’ve saved, what’s due for review, and what you’ve been struggling with recently — not randomly.

It Knows You — The Learner Profile

The first time you start chatting, the AI will ask you a few quick questions: - What is your current level? - What is your native language? - What is your daily study goal?

These answers build your **Learner Profile**, which the AI reads at the start of every session to calibrate the difficulty of its explanations and exercise selections. You can update it at any time simply by telling the AI — “I think I’m at intermediate level now” — and it will adjust.

Long-Term Memory

The AI keeps a **persistent memory** of important things you share with it: - Your learning goals and motivations - Topics or grammar patterns you find difficult - Preferences for how you like to be taught - Observations it makes about your performance over time

This memory carries over to every future session. The AI can also search its own memory when you ask — “*What have I told you about my goals?*” — and retrieve what it knows about you.

Ask for Your Progress

At any time, type: - “*Show me my progress chart.*” - “*How is my skill breakdown looking?*”

The AI will render a **visual chart** right in the chat — an accuracy trend graph or a mastery breakdown across all five skills (Listening, Speaking, Reading, Writing, Vocabulary).

Tips for Getting the Best Out of It

- **Be specific.** “*Quiz me on 想 and 觉得*” works better than “*quiz me*”.
- **Tell it your mood.** “*I only have 5 minutes*” → it will keep the session short.
- **Ask it to explain differently.** If an explanation doesn’t click, say “*explain that another way*”.
- **Trust its suggestions.** When it says “*you have 3 words due for review*”, take the quiz — spaced repetition works best when you don’t skip.

6. Speak with AI — Live Voice Practice

No scheduling. No judgment. Just you and an AI that corrects your grammar while you speak.

Speak with AI is a real-time voice conversation feature. You speak — the AI responds in your target language — and the app gives you instant support so you never get stuck. Before using this feature, make sure you’ve added your **Google Gemini API key** in Settings (see [Initial Setup ↑](#)).

What You Get in Every Session

Feature	What it does
Voice conversation	Speak freely; AI responds naturally in your target language
Grammar feedback	Every message silently analyzed — tap it to see corrections
Next-Line Suggestion	Can’t think of what to say? AI suggests your next line

Feature	What it does
Cultural tips	Notes on idioms, formality, and cultural nuance
Vocabulary checklist	Target words auto-ticked as you use them
Session recap	Full summary with all grammar corrections and an overall evaluation

Supported languages: Chinese (Simplified), English, Japanese. **Session limit:** 10 minutes per session (countdown timer shown during conversation).

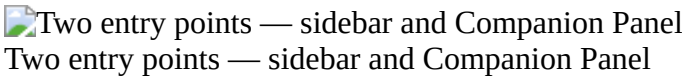
How to Use It

Step 1 — Open Speak with AI

There are two ways to access the feature:

- **From the Topbar:** Click the **Speak with AI** button in the left navigation menu.
- **From the Companion Panel:** Inside a lesson, click the **Voice** icon in the AI Companion panel on the right.

If your Google Gemini key isn’t set yet, the app will prompt you to go to Settings first.

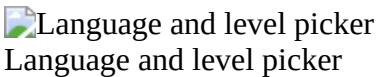


Step 2 — Choose language and level

Select your **target language**, then pick a **proficiency level**:

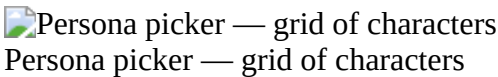
- **Beginner** — HSK 1-2 / CEFR A1-A2 / JLPT N5-N4 / TOPIK I
- **Intermediate** — HSK 3-4 / CEFR B1-B2 / JLPT N3-N2 / TOPIK II 3-4
- **Advanced** — HSK 5-6 / CEFR C1-C2 / JLPT N1 / TOPIK II 5-6

The AI adjusts vocabulary, sentence complexity, and feedback depth to match your level. Click **Continue**.



Step 3 — Pick a persona

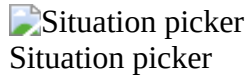
Choose the AI character you want to talk to. Each persona has a distinct personality and speaking style — friendly study buddy, encouraging friend, strict parent, taxi driver, and more. Available personas vary by language.



Step 4 — Pick a situation

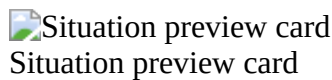
Choose a scenario from the grid: casual chat, ordering food, asking directions, shopping/bargaining, job interview, meeting the family, hospital visit, karaoke, and more.

Can't find the right scenario? Click **** Create your own**** at the top of the grid. Describe your situation in plain text and the AI generates a custom scenario with matching vocabulary.



Step 5 — Preview and confirm

Before starting, review the scenario card: description, your role and goal, and the target vocabulary for this session. Click **Regenerate** for a fresh version, or **Start** when ready.



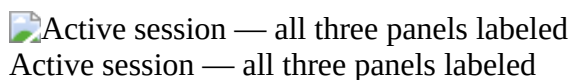
Step 6 — Have the conversation

The session screen has three panels:

Center — Conversation - AI avatar with a mic ring that pulses when the AI is speaking - Live transcript of the full conversation - **Hold microphone button to speak, release when done** - Session countdown timer from 10:00

Right — Intelligence Panel - Target vocabulary checklist — words get crossed off as you use them - **Next-Line Suggestion** — a contextual prompt to help continue the conversation - **Cultural tips** — idioms, register, situational etiquette - **Vocab tips** — quick explanations of words the AI just used

Left — Grammar Feedback - Click any message in the transcript to open its grammar panel - Corrections shown as: **original** → **corrected**, with explanation



Tips: Speak at a natural pace. If you go silent, the AI gently prompts you. Use Next-Line Suggestion when stuck — say it aloud and see how the AI responds. Grammar corrections appear automatically.

When finished, click **End Session**.

Step 7 — Review your recap

After the session, a full recap shows: - Duration and turn count - Overall AI evaluation of your performance - Strengths and areas to improve - Grammar correction timeline — every correction from the session, in order

Click **Practice Again** to repeat the same scenario, or **Back** to choose a new one.



Session recap

Troubleshooting

- “Google API key required” error → Go to Settings and add your Google Gemini API key.
- Microphone not working → In Chrome/Edge: click the lock icon in the address bar → allow microphone.
- Session won’t start / loading spinner hangs → Check your internet connection. The session requires a stable connection to the voice server.
- Choppy or delayed audio → Close other tabs using audio. Move closer to your router or switch to a wired connection.

7. Shadowing Practice (Pronunciation & Listening)

Click the **Shadowing** button in the middle column (or the icon on a specific segment). A dialog will appear where you choose your mode and how many segments to practice.

Choosing a Mode



Mode	What You Do
Dictation Mode	Listen to a segment and type what you hear
Speaking Mode	Listen to a segment and speak it aloud into your microphone

You can also set **how many segments** to practice, or choose “All remaining” to go through the whole lesson.

Dictation Mode Flow

- Step 1 — Listen Phase** - The segment audio plays automatically. - Focus on the sounds, rhythm, and tone. Use the **Replay** button or press **Space** to hear it again.
- Step 2 — Type Phase** - A text input appears. Type exactly what you heard. - Click **Submit** when done.
- Step 3 — Reveal Phase** - Your answer is shown alongside the correct text. - Each character is highlighted: Green = correct, Red = wrong. - Your accuracy score is shown. - Click **Retry** to try again or **Next** to move on.



Speaking Mode Flow

Step 1 — Listen Phase - The segment audio plays automatically. Press **Space** or **Replay** to hear it again.

Step 2 — Speak Phase - Click the **microphone button** (or press **Space**) to start recording. - Speak the segment aloud clearly, then click again to stop. - You can play back your recording before submitting, or click **Redo** to re-record. - Click **Submit** when satisfied.

Step 3 — Reveal Phase - Your recording is played back alongside the correct AI pronunciation. - Azure grades your pronunciation: Green = good, Red = needs work. - Click **Retry** to try again or **Next** to continue.

Shadowing Session Summary

After all segments are done, you see a full summary: - Accuracy score per segment - Which segments you attempted, skipped, or were auto-skipped (very short segments are skipped automatically) - Overall session average score

8. Vocabulary Workbook, Quizzes & Progress

Saving Words

While reading subtitles, **hover over any word** and click the **Bookmark** icon. The word is instantly added to your Workbook.

Accessing the Full Workbook

Click **Workbook** in the top navigation to open the full Vocabulary page. Here you can:

- See **total words saved** and how many lessons they come from
- **Search** for any word by character, pinyin, or meaning
- Browse words **grouped by lesson** — click a group to expand or collapse it
- Click **Play** on any word to hear it pronounced
- Click **Study** on any lesson group to practice only those words



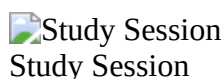
Quiz Formats

When you start a study session (from the Workbook or the lesson's Workbook tab), choose your exercise type and how many words to practice:

Quiz Mode	What You Do	Skill Trained
Mixed	Randomly cycles through all exercise types	All-round practice

Quiz Mode	What You Do	Skill Trained
Cloze	Fill in blanks in an AI-generated story	Reading in context
Dictation	Listen and type the Chinese sentence	Listening & spelling
Pronunciation	Read the sentence aloud for Azure to grade	Speaking accuracy
Reconstruction	Arrange jumbled words into the correct sentence	Grammar & word order
Pinyin Recall	See the character — can you recall its pinyin?	Reading & pronunciation
Character Writing	Trace the stroke order of each character on screen	Writing & muscle memory

Each exercise has an ⓘ **info button** — click it to read a description of the exercise before starting.



Study Session
Study Session

After finishing, a **Session Summary** shows your score breakdown by exercise type.

Progress Dashboard

The bottom of the Workbook page shows a live progress dashboard that updates automatically as you complete exercises.

Overall Stats

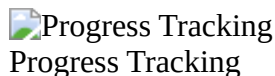
Displays your total exercise count, overall accuracy rate, and total study time accumulated across all sessions.

Accuracy Trend

A chart showing your daily accuracy over the past 90 days — one data point per day. It reflects the running accuracy of all exercises you answered that day.

Skill Breakdown

Five cards — **Writing, Speaking, Vocabulary, Reading, Listening** — each showing: - Accuracy percentage (correct answers ÷ total attempts for that skill) - Total exercise attempts logged



Progress Tracking
Progress Tracking

Each exercise type feeds into a specific skill:

Exercise	Skill tracked
Character Writing	Writing
Translation	Writing
Pronunciation	Speaking
Pinyin Recall	Speaking
Cloze	Vocabulary
Reconstruction	Reading
Dictation	Listening

How scoring works: Binary exercises (Dictation, Cloze, Reconstruction, Pinyin Recall) score 100 for correct or 0 for wrong. Character Writing scores 100 if no hints were used, 80 if hints were used. Translation and Pronunciation receive real numeric scores (0–100) from AI. A score of **60 or above** counts as a correct answer for accuracy purposes.

Common Mistakes

Highlights vocabulary items you have struggled with most frequently, so you know where to focus.

Spaced Repetition & Review Queue

Every time you answer an exercise, ShadowLearn schedules the next review for that vocabulary word using a scientifically-proven method that spaces reviews over increasing intervals based on how well you know the word:

- Answer correctly and confidently → next review in several days
- Answer incorrectly → reviewed again soon (tomorrow or the day after)
- Consistently correct over 5 sessions → the word levels up and is reviewed less often

When words are due for review, a banner appears at the top of the Workbook page: **“X items due for review today”**. Click **Start Review** to launch a targeted session with only those words.

9. Settings

Click the **Settings icon** () in the top navigation at any time to:

- **Update API Keys** — paste new or corrected keys; use the icon to show/hide a key
- **Change PIN** — enter a new PIN and confirm it
- **Set Default Translation Language** — choose which language lessons translate into by default
- **Lock App** — immediately lock the app (requires PIN to re-open)

Forgot PIN button: This resets the app and erases all stored API keys. You will need to re-enter them on next launch. Use only as a last resort.

10. Frequently Asked Questions (FAQ)

Account & Security

Q: Can I change my PIN? - A: Yes — open Settings and use the **Change PIN** section. Write your PIN down somewhere safe; there is no recovery if forgotten.

Q: Where is my data stored? - A: Everything — lessons, vocabulary, chat history, AI memory, audio cache — is stored locally in your browser's storage. Nothing is sent to our servers.

Lessons & Library

Q: How long does YouTube lesson creation take? - A: Usually 1–3 minutes depending on video length (download + transcription + translation).

Q: The lesson shows an error. What do I do? - A: For YouTube lessons, click **Retry** on the lesson card in the Library. For uploaded files, delete the lesson and upload again.

AI Companion & Memory

Q: The AI Companion doesn't seem to know about my lesson / vocabulary. - A: The AI looks up your data when it needs it (when you ask for a quiz or progress check). Just ask directly — “*quiz me on my saved words*” — and it will fetch them.

Q: Can the AI Companion remember things between sessions? - A: Yes. It has a long-term memory system. Tell it your goals, level, or problem areas and it will remember for future sessions.

Shadowing & Exercises

Q: Why are some segments skipped in Shadowing mode? - A: Very short segments (fewer than 3 words) are automatically skipped — they are too brief to practice meaningfully.

Wishing you productive and enjoyable learning sessions on ShadowLearn!